

Top 10 German Technology Companies

Building Industrial AI, Physical AI and Agentic Commerce

How Germany can combine AI agents, industrial software, manufacturing and IoT assets, autonomous machines and European trust infrastructure to create a globally competitive agentic cyber-physical economy.

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THE 2030 PROPOSITION

Germany already has the components. Its strongest path is a coordinated, open and globally interoperable industrial stack in which capable AI agents and machines can act with verifiable authority, reliable evidence and clear accountability.

Abstract

This is a strategic selection of Germany's most advanced technology companies for agentic AI, Industrial AI, Physical AI and agentic commerce. It examines how Germany can combine AI agents, industrial software, manufacturing and IoT assets, autonomous machines and European trust infrastructure to create a globally competitive agentic cyber-physical economy.

Keywords. Industrial AI, Physical AI, agentic commerce, cyber-physical systems, trust infrastructure, Germany

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METHODOLOGY NOTE

This is a strategic selection of Germany's most advanced technology companies for agentic AI, Industrial AI, Physical AI and agentic commerce.

1 Executive answer: Germany has the components, now it needs the trusted system

Germany already has many of the companies needed to build the agentic cyber-physical economy: AI developers, workflow and process platforms, autonomous systems, robotics, energy orchestration, chips, cloud and industrial automation. The missing strategic layer is shared trust infrastructure that lets organizations, products, machines and AI agents prove identity, authority, evidence and accountability across company and national borders.

This article selects ten German technology companies for their strategic relevance rather than simply their size or latest valuation. Presented in alphabetical order:

1. **1KOMMA5° / Heartbeat AI** — distributed energy orchestration
2. **Black Forest Labs** — frontier visual AI models
3. **Celonis** — process intelligence and enterprise digital twins
4. **Helsing** — defence AI and autonomous mission systems
5. **n8n** — AI workflow and agent orchestration
6. **NEURA Robotics** — cognitive and humanoid robotics
7. **Parloa** — enterprise conversational AI agents
8. **Quantum Systems** — software-defined autonomous systems
9. **Sereact** — Physical AI for robotic manipulation
10. **Spherity** — organizational, product and agent trust infrastructure

The selection highlights the route from AI capability to industrial-scale outcomes. Germany's next productivity wave will not come from models alone. It will emerge when agents can interpret reliable industrial evidence, control physical systems and enter governed transactions. That includes ordering components, procuring energy, booking logistics, contracting services and exchanging regulated data.

This paper examines one practical question for scaling AI into operational systems through strategic, globally oriented cooperation:

How can every relevant counterparty verify that an AI agent acting in the physical economy is properly identified and authorized, relies on trustworthy evidence, remains within cybersecurity, risk, compliance and functional-safety constraints, and leaves an auditable record of responsibility before, during and after the action?

This is where Spherity occupies a strategically complementary position alongside the other nine companies and Germany's established technology anchors. It does not need to replace their AI, robotics, cloud, semiconductor, automation or commerce platforms. Instead, Spherity can provide a cross-company trust layer connecting identities, mandates, product and AI-system evidence, runtime authorization and auditable records. This can reduce verification and compliance friction while enabling the secure automation of sensitive B2B, B2G and cross-border

processes, including critical infrastructure and civil–military or dual-use supply chains.

Europe’s opportunity lies in combining such trust infrastructure with an open, model-agnostic and globally interoperable foundation for Industrial AI, Physical AI and agentic commerce. Its competitive advantage will not come from any single model or platform, but from the ability to deploy autonomous systems at scale with demonstrable cybersecurity, functional safety, compliance and accountability. This can turn Europe’s regulatory and engineering strengths into an exportable architecture for the global physical economy.

2 Deutsche Zusammenfassung

Deutschland verfügt bereits über wesentliche Bausteine einer agentenbasierten cyber-physischen Wirtschaft: Halbleiter, souveräne Cloud-Infrastruktur, Unternehmenssoftware, industrielle Daten, KI-Modelle, Robotik, autonome Systeme und Energietechnologie. Der nächste Wettbewerbsvorteil entsteht, wenn diese Fähigkeiten über Unternehmens- und Ländergrenzen hinweg vertrauenswürdig zusammenarbeiten können.

Entscheidend ist nicht nur, ob ein KI-Agent eine Aufgabe technisch ausführen kann. Geschäftspartner müssen prüfen können, welche Organisation der Agent vertritt, welche Befugnisse er besitzt, welche Produkt- und Compliance-Nachweise zugrunde liegen und wer für die Handlung verantwortlich bleibt. Zahlungen allein lösen diese Fragen nicht.

Die ausgewählten zehn Unternehmen—1KOMMA5°, Black Forest Labs, Celonis, Helsing, n8n, NEURA Robotics, Parloa, Quantum Systems, Sereact und Spherity—bilden unterschiedliche Teile dieses Systems ab. Große Industrie- und Technologieunternehmen wie Siemens, Bosch, SAP, Infineon, Deutsche Telekom, Schwarz Digits, Schaeffler, Phoenix Contact, Festo und TRUMPF bringen Produktionsumgebungen, Infrastruktur und globale Marktzugänge ein.

Dieses Paper untersucht eine zentrale Frage für die Skalierung von KI in operativen Systemen:

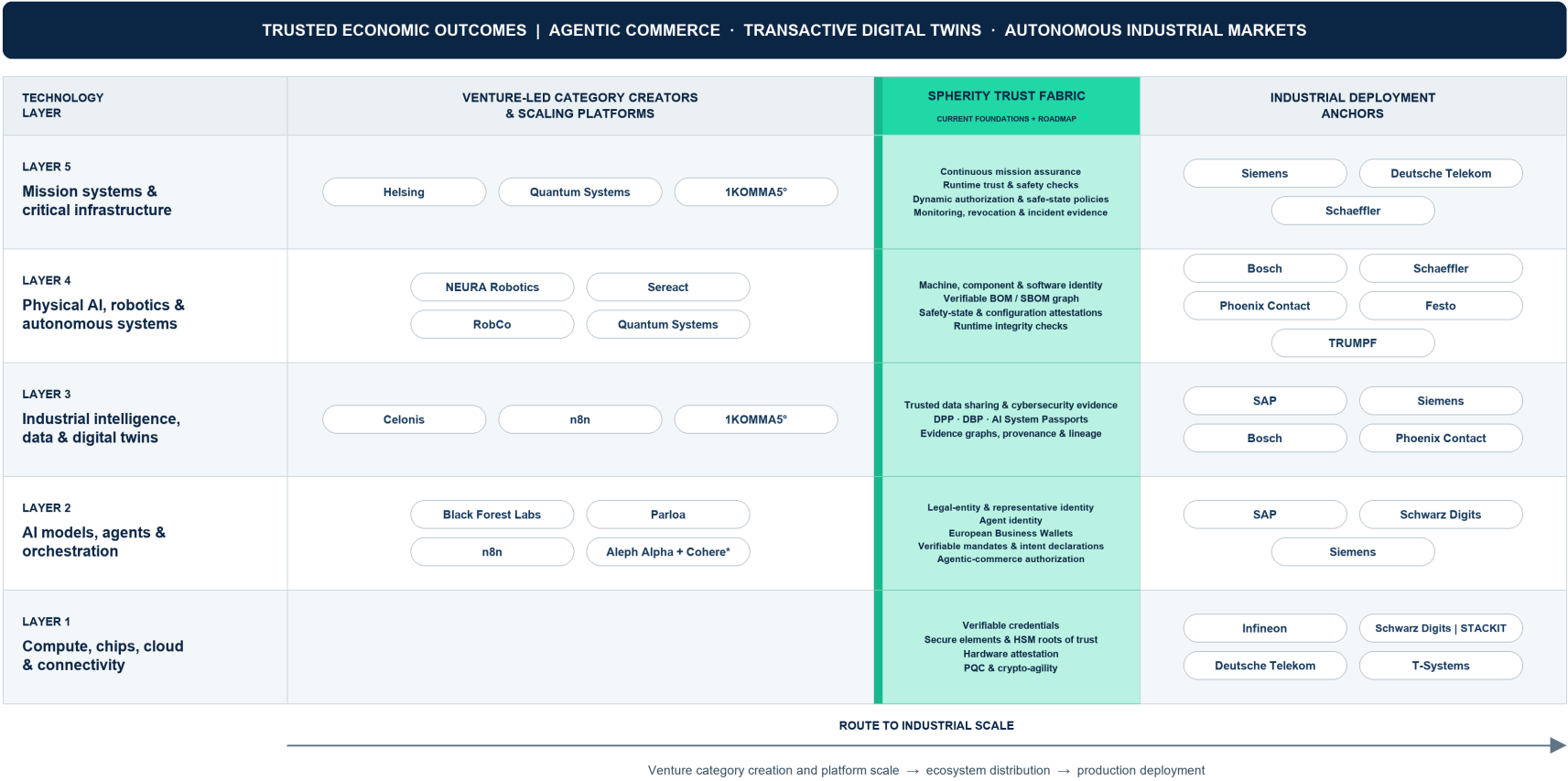
Wie kann jeder relevante Geschäftspartner überprüfen, dass ein KI-Agent in der physischen Wirtschaft eindeutig identifiziert und rechtmäßig autorisiert ist, auf vertrauenswürdigen Nachweisen basiert, Cybersecurity-, Risiko-, Compliance- und funktionale Sicherheitsanforderungen einhält und vor, während sowie nach der Handlung eine auditierbare Verantwortungs- und Nachweiskette hinterlässt?

Hier liegt die strategisch komplementäre Position von Spherity gegenüber den neun anderen Unternehmen und den etablierten deutschen Technologieankern. Spherity muss deren KI-, Robotik-, Cloud-, Halbleiter-, Automatisierungs- oder Handelsplattformen nicht ersetzen. Stattdessen kann Spherity eine unternehmensübergreifende Vertrauensebene bereitstellen, die Identitäten, Mandate, Produkt- und KI-Systemnachweise, PQC-fähige Vertrauensketten, Laufzeitautorisierungen und auditierbare Nachweise verbindet. Dadurch lassen sich Prüf- und Compliance-Aufwände reduzieren und sensible B2B-, B2G- und grenzüberschreitende Prozesse sicher automatisieren, einschließlich kritischer Infrastrukturen sowie zivil-militärischer und Dual-Use-Lieferketten.

Europas Chance liegt darin, eine solche Vertrauensinfrastruktur mit einer offenen, modellunabhängigen und global interoperablen Grundlage für industrielle KI, Physical AI und Agentic Commerce zu verbinden. Der Wettbewerbsvorteil entsteht nicht durch ein einzelnes Modell oder eine einzelne Plattform, sondern durch die Fähigkeit, autonome Systeme nachweisbar cybersicher, funktional sicher, regelkonform und verantwortlich zu skalieren. So können Europas regulatorische und ingenieurtechnische Stärken zu einer exportierbaren Architektur für die globale physische Wirtschaft werden.

Germany's Agentic Cyber-Physical Technology Stack

A coordinated route from venture-led innovation to trusted industrial deployment



* Planned Cohere-Aleph Alpha combination; closing not confirmed at the 2 August 2026 research cut-off.

Research synthesis: Spherity · 2026

Figure 1: Germany’s Agentic Cyber-Physical Technology Stack. Five technology layers map German ventures, scaling platforms and industrial anchors. An uninterrupted Spherity Trust Fabric crosses all layers, progressing from verifiable credentials and hardware-rooted trust through organizational and agent authority, trusted evidence, runtime assurance and continuous mission assurance. The column distinguishes current foundations from strategic roadmap capabilities. Germany’s venture-led technologies and corporate anchors supply complementary capabilities; certified systems retain responsibility for physical control. Source: Spherity research; company and regulatory facts checked to 2 August 2026.

3 Selection method: strategic relevance

The Top 10 is an editorial selection for the emerging agentic industrial economy. Companies were assessed across six dimensions: differentiated technology; strategic market importance; evidence of commercial scale or rapid adoption; relevance to Industrial AI, Physical AI or agentic commerce; European sovereignty and resilience; and potential to create complementary value with organizational, product and agent trust infrastructure.

Funding and valuation are useful evidence of market confidence and the capacity to execute, but they are not proxies for technology quality. The list also deliberately mixes established scale-ups, fast-growing ventures and a specialized trust-infrastructure company. Established German corporations are treated separately as deployment anchors because their role is different: they provide factories, infrastructure, distribution, standards influence and installed customer bases.

4 The five-layer agentic cyber-physical technology stack

Germany's opportunity becomes clearer when its companies are viewed as a system rather than as isolated success stories.

Layer 1—compute, chips, cloud and connectivity. Infineon supplies semiconductors, sensing, edge intelligence and hardware security. Schwarz Digits and its STACKIT cloud, together with Deutsche Telekom and T-Systems, provide sovereign infrastructure and connectivity. This layer gives industrial AI a place to compute and a secure route to exchange information. Spherity's current foundations include verifiable credentials and crypto-agility; its roadmap connects them to secure-element and HSM roots of trust, hardware attestation and PQC migration.

Layer 2—AI models, agents and orchestration. Black Forest Labs develops visual foundation models. Parloa deploys enterprise customer-service agents. n8n coordinates models, tools and workflows. The planned Cohere–Aleph Alpha combination aims to strengthen sovereign, enterprise-grade AI. This layer interprets information and turns goals into planned actions. Spherity can connect legal-entity, representative and agent identity through European Business Wallets, verifiable mandates, signed intent declarations and agentic-commerce authorization.

Layer 3—industrial intelligence, data and digital twins. Celonis models how business processes actually run. SAP embeds enterprise and manufacturing data into operational workflows. Siemens and Bosch connect engineering, factory and lifecycle information. This layer supplies the context needed to make an agent's output useful in a real production system. Spherity can provide trusted data-sharing and cybersecurity evidence through Digital Product Passports, Digital Battery Passports, future AI System Passports, evidence graphs, provenance and lineage.

Layer 4—Physical AI, robotics and autonomous systems. NEURA Robotics, Sereact, Quantum Systems and RobCo turn perception and reasoning into movement. Bosch, Schaeffler, Phoenix Contact, Festo and TRUMPF bring components, controls, safety, production knowledge and industrial distribution. This layer acts on the physical world. Spherity's strategic roadmap adds machine, component and software identity; verifiable BOM and SBOM graphs; safety-state and configuration attestations; and runtime integrity checks around—not in place of—certified controllers.

Layer 5—mission systems and critical infrastructure. Helsing, Quantum Systems and 1KOMMA5° operate in environments where reliability, security and real-world consequences matter. Siemens, Deutsche Telekom and Schaeffler anchor these capabilities in industrial and infrastructure markets. Spherity's longer-term role is continuous mission assurance: runtime

trust and safety checks, dynamic authorization and safe-state policies, plus monitoring, revocation and incident evidence. Certified safety and operational systems remain responsible for executing physical control actions.

Above the stack are the economic outcomes: **agentic commerce, transactive digital twins and autonomous industrial markets**. These are not a sixth technology layer. They result when the other layers can interact with verifiable authority, evidence and accountability.

Spherity's role is therefore shown as a vertical trust fabric across all five layers. The five layers answer *what the system can do*. The trust fabric answers *how trust is anchored, who or what may act, which evidence can be relied upon, whether systems remain in an approved state and how consequential actions can be controlled and audited*.

The chart deliberately combines **current foundations and a strategic roadmap**. Spherity's present strengths are most mature in organizational identity, wallets, verifiable credentials, regulated data exchange and product evidence. Hardware attestation, AI System Passports, continuous runtime assurance and safe-state policy orchestration are adjacent roadmap capabilities to be developed with semiconductor, automation, functional-safety, cybersecurity and mission-system partners. This distinction preserves claims discipline while showing how the trust fabric can expand as cyber-physical autonomy matures.

No single company can build the entire system. A cloud provider cannot by itself establish a supplier's legal authority. A payment confirms that funds moved, not that a component met regulatory requirements. A Digital Product Passport can hold useful evidence, but it does not automatically grant an AI agent permission to purchase. An accurate model does not prove that its operator was entitled to make a binding commitment. Industrial autonomy therefore requires a federation of specialized capabilities.

5 Agentic commerce, Industrial AI and Physical AI: from recommendation to governed transaction

Agentic commerce is commerce in which AI agents discover products and services, evaluate evidence, compare alternatives, negotiate terms, place orders, authorize transactions, coordinate fulfilment and document outcomes on behalf of companies, people, machines or public authorities.

It includes consumer assistants, enterprise procurement, industrial machine-to-machine commerce, energy and flexibility trading, automated maintenance, B2B2C services, and B2G or G2G processes. It extends well beyond checkout. Open payment and commerce protocols are making it easier for agents to interact with merchant and payment systems. Yet regulated industry needs additional assurance about identity, representation, product provenance, eligibility, authorization, revocation and audit evidence.

CORE FORMULATION

Agentic commerce cannot scale on identity assumptions. Before an AI agent can order a component, procure energy, negotiate a service or initiate a regulated transaction, counterparties must be able to verify whom the agent represents, what it is permitted to do, which product and compliance evidence it evaluated and who remains accountable.

Four maturity stages help separate useful automation from trusted autonomy:

1. **Observe.** AI reads operational, product and market data.

2. **Recommend.** AI proposes a decision to a human.
3. **Act.** An authorized agent executes an approved operational action within a system.
4. **Transact.** Agents negotiate or exchange products, services, energy, data or capacity across organizations.

The fourth stage is materially harder because a transaction crosses governance boundaries. A manufacturer can configure its own workflow internally. A supplier, bank, regulator or public authority cannot simply accept that internal configuration as proof of authority. Cross-company autonomy needs verifiable credentials, bounded mandates, status and revocation checks, product or asset evidence, policy enforcement and durable action receipts.

5.1 Two trust planes for Industrial AI & Physical AI

Spherity's architecture can be explained through two connected planes.

The **control plane** answers: *Who is authorized to act, under which conditions, and what should happen when those conditions change?* It covers legal-entity, representative and agent identity; delegated mandates; signed intent declarations; spending and operational limits; permitted suppliers and jurisdictions; runtime policy decisions; credential status; revocation; escalation; safe-state policies; and receipts showing what was authorized and executed. In safety-critical environments, Spherity can supply verifiable policy inputs and evidence while certified operational technology executes the physical control action.

The **data plane** answers: *What evidence supports the action and the current system state?* It covers provenance, validation, freshness, product and asset evidence, DPPs, DBPs and future AI System Passports (AISP), BOM and SBOM relationships, configuration and safety-state attestations, transformations, conflicting information, uncertainty, issuer status and the evidence path used by the agent.

TWO-PLANE PRINCIPLE

The distinction is essential: A valid mandate does not prove that an AI decision is factually supported or operationally safe. Reliable evidence does not prove that the AI agent was legally authorized to act. Regulated Industrial AI, Physical AI and trusted agentic commerce require both. Agentic transactions emerge when the two planes connect to transaction execution; continuous mission assurance emerges when they also connect to monitoring, runtime checks, revocation, incident evidence and approved control responses.

6 Germany's corporate technology anchors

Germany's large technology and industrial groups are not supporting characters. They are the route from promising technology into plants, supply chains, regulated markets and global customer networks.

Siemens links industrial software, automation, digital twins and AI agents. Its 2026 industrial AI announcements included an engineering agent and broader industrial copilots. Siemens can make agentic systems operational across engineering and manufacturing; Spherity can add verifiable supplier authority, product evidence and cross-company action receipts.

Bosch combines sensors, control systems, manufacturing and robotics. Its Physical AI activities and partnership with NEURA Robotics illustrate how factory knowledge and robot capability

reinforce each other. Component identity, training-data provenance and authorized maintenance actions are natural trust extensions.

SAP is the enterprise transaction and data backbone for much of global industry. Its manufacturing and supply-chain agents can turn enterprise context into action. Spherity can help make external credentials, mandates and evidence portable across company boundaries rather than trapping them inside a single tenant or platform.

Infineon anchors the lowest layer of the stack: chips, sensing, edge AI and hardware security. Chip-rooted device identity can let machines and components authenticate before exchanging data or participating in transactions. Combined with Spherity credentials and crypto-agility, this creates an end-to-end chain from secure silicon to organizational authorization and product provenance.

Deutsche Telekom and T-Systems provide connectivity, sovereign cloud and a production route for regulated industries. Their Industrial AI Cloud, launched in 2026 with roughly 10,000 GPUs, is a national-scale deployment asset. Trust services can help its agents and data services interoperate beyond one cloud boundary.

Schwarz Digits is a sovereign-infrastructure anchor, not a conventional venture-backed scale-up. Its portfolio spans STACKIT cloud, cybersecurity, AI, communication and workplace services; the business reported €2.2 billion in fiscal 2025 revenue. It is also central to the planned Cohere–Aleph Alpha combination. Its strategic importance lies in bringing sovereign compute, enterprise distribution and regulated AI together.

Schaeffler connects industrial components, actuators, factories, robotics and lifecycle data. Its NEURA collaboration includes humanoid deployment, component supply and data generation for Physical AI. Verifiable component passports, robot mandates and maintenance evidence could allow machines to request services or replacements within predefined limits.

Phoenix Contact brings industrial controls, PLCnext, edge computing, functional safety and IEC 62443 cybersecurity. Its authenticated controllers could constrain which AI agents may change a machine configuration, install software or initiate a purchase. Spherity’s authorization and receipt layer would complement safety-certified control rather than bypass it.

Festo contributes automation components, intelligent handling, robotics and industrial skills. Its 2026 GripperAI launch shows how flexible perception is moving into production tooling. Verified end-effectors, training evidence and technician credentials could reduce integration and service friction.

TRUMPF combines machine tools, lasers, production software and a global industrial customer base. Its equipment can become a trusted participant in service, spare-parts and capacity markets when machine identity and lifecycle evidence are portable.

6.1 Sovereign AI consolidation: Aleph Alpha, Cohere and Schwarz Digits

On 24 April 2026, Canada’s Cohere and Germany’s Aleph Alpha announced a planned combination to create a sovereign enterprise AI provider, supported by a €500 million structured-financing commitment from the companies of Schwarz Group. As of this article’s 2 August 2026 research cut-off, the latest authoritative language reviewed still described an intent to join forces; it should not be presented as a completed merger without confirmation of closing.

Aleph Alpha remains strategically important because it has focused on transparent, controllable AI for public-sector and regulated enterprise environments. The proposed transaction changes how it should be classified: it is no longer best treated as an independent German scale-up in a Top 10 selection. Instead, the combination belongs in the scaling-platform column of the

matrix, with Schwarz Digits and STACKIT as infrastructure and distribution anchors.

Even a sovereign model running on sovereign cloud does not automatically provide cross-company authority and evidence. Models need to know which organization an agent represents, whether its mandate is valid, which product claims are current and whether a counterparty may rely on them. This creates a complementary role for standards-based trust infrastructure.

7 The Top 10 German technology companies (in alphabetical order)

7.1 1KOMMA5° / Heartbeat AI: distributed energy becomes transactive

1KOMMA5° integrates residential energy systems—solar, storage, heat pumps and electric-vehicle charging—with its Heartbeat AI platform. Its virtual power plant coordinates distributed assets and uses market signals to optimize consumption and trading. The company targets 500 megawatts of controlled capacity in 2026.

Its defensible position combines customer access, installed hardware, software and energy-market integration. The catalyst is electrification, volatile power prices and the need to coordinate flexible distributed assets. Risks include hardware-heavy execution, policy and tariff changes, heterogeneous devices and the operational responsibility of controlling energy assets.

This is a direct path to machine-to-market commerce. A battery agent can verify asset identity, capacity, state of health, ownership and grid permission before offering flexibility. Spherity could add asset passports, installer credentials, manufacturer evidence, delegated trading mandates and durable receipts across household, aggregator, grid and market participants.

7.2 Black Forest Labs: visual intelligence for digital and physical systems

Freiburg-based Black Forest Labs develops the FLUX family of image-generation and visual AI models. Its December 2025 Series B announcement reported \$300 million raised at a \$3.25 billion post-money valuation. The company’s defensibility rests on frontier model expertise, adoption in developer ecosystems and the ability to make visual generation controllable and deployable.

The catalyst is growing demand for multimodal systems that understand and create visual content for media, design, simulation and industrial workflows. Risks include rapid model competition, compute intensity, copyright and provenance questions, and the challenge of translating broad adoption into durable enterprise economics.

Its agentic-commerce relevance is indirect but important. Visual agents can generate product content, inspect assets or interpret imagery. Spherity could add model and organizational identity, AI System Passport evidence, provenance credentials, usage-policy evidence and receipts so that enterprises know which model, mandate and evidence chain produced a commercial output.

7.3 Celonis: process intelligence as the enterprise context layer

Celonis turns event data into a view of how business processes actually operate. Founded in Munich in 2011, it reports more than 3,000 employees and over 1,400 customers. Its Process Intelligence Graph functions as a system-agnostic business context layer—effectively a living

model of objects, events and relationships across enterprise processes.

The defensible advantage is not just process mining; it is the accumulated integration, process knowledge and operational deployment needed to improve large enterprises. AI agents need that context to avoid optimizing one step while damaging the wider system. The catalyst is enterprise demand for agents that can move from chat to measurable execution. The risk is platform competition and the difficulty of proving sustained value across complex transformations.

In agentic commerce, Celonis can detect the need for an action and orchestrate the process around it. Spherity can supply external organizational authority, supplier credentials, product evidence and receipts. Together, a process agent could identify a supply bottleneck, verify eligible suppliers and initiate a governed procurement workflow.

7.4 Helsing: software-defined defence and autonomous mission systems

Helsing develops AI and autonomous systems for defence. In July 2026 it announced a \$1.8 billion Series E at an \$18 billion valuation, evidence of its strategic importance and execution capacity rather than proof of product superiority. Its defensible position combines operational software, domain data, integration with platforms and a predominantly European ownership base.

The market catalyst is Europe's move toward greater defence capability, autonomous systems and software-defined procurement. The risk is equally clear: programmes have long qualification cycles, high assurance requirements and geopolitical exposure. Systems must work in contested environments where provenance, supply-chain integrity and cryptographic resilience are mission requirements.

Helsing's agentic-commerce relevance lies in secure supplier, maintenance, logistics and mission-support corridors rather than consumer transactions. Spherity could complement it with PQC-ready organizational and supplier credentials, mission-data entitlements, component evidence and auditable delegation across allied ecosystems.

7.5 n8n: the orchestration layer for enterprise agents

Berlin-based n8n provides a workflow automation platform that connects applications, data and AI models. Its October 2025 Series C announcement reported \$180 million raised, \$240 million in total funding and a \$2.5 billion valuation. Its most defensible feature is flexible, developer-friendly orchestration that can run in controlled environments while connecting a wide range of systems.

The catalyst is the shift from isolated AI assistants to agents that call tools and complete workflows. The risk is intense competition from cloud, enterprise-software and open-source automation providers, together with the governance burden created when workflows gain transactional authority.

n8n can orchestrate the steps of agentic commerce, but orchestration is not authorization. An n8n workflow using Spherity could check an agent credential and mandate, validate supplier and product evidence, enforce a spending policy, record the action receipt and then call the relevant procurement or payment service.

7.6 NEURA Robotics: a German full-stack Physical AI contender

NEURA Robotics develops cognitive robots and humanoid systems. In June 2026 it announced a Series C of up to \$1.4 billion and described a full-stack Physical AI strategy built around robots, sensors, control, AI and the Neuraverse ecosystem. Partners and investors named by the company include major semiconductor, cloud and industrial groups.

Its opportunity is to make adaptable robots usable beyond fenced, repetitively programmed cells. The defensible asset is integration across hardware and software, reinforced by ecosystem partnerships and industrial data. The catalyst is labour scarcity and demand for flexible automation. The principal risk is turning ambitious humanoid roadmaps into reliable, safe, cost-effective deployments at scale.

For agentic commerce, a robot could diagnose wear, verify an approved component, request maintenance and procure within a defined budget. Spherity could provide robot and organization identity, component passports, technician credentials, training and safety evidence, mandates and auditable receipts.

7.7 Parloa: enterprise customer interactions become agentic

Parloa develops AI agents for enterprise customer service. In January 2026 it announced a \$350 million Series D at a \$3 billion valuation. Its strategic advantage is specialization in enterprise-grade voice and conversational interaction, where integration, latency, language quality, compliance and operational reliability matter.

The catalyst is the replacement of scripted contact-centre automation with agents that can reason across tools and customer context. Risks include model commoditization, errors in high-stakes customer interactions and regulatory exposure when an agent makes commitments.

Parloa is close to consumer and B2B agentic commerce: an agent may change a service, issue a refund, place an order or negotiate retention terms. Spherity can help verify the enterprise behind the agent, bind delegated authority to transaction limits, validate relevant customer or product credentials and generate action receipts that support audit and dispute handling.

7.8 Quantum Systems: autonomous systems across air, land and sea

Quantum Systems develops software-defined autonomous systems and unmanned platforms. In July 2026 it announced a \$1.2 billion Series D at an approximately \$8 billion post-money valuation. The company emphasizes a multi-domain architecture across air, land and sea and says it has demonstrated profitability.

Its advantage is the integration of hardware, autonomy software, sensing and mission workflows. Demand for European defence technology and dual-use autonomy is a strong catalyst. Execution risks include production scale, procurement concentration, export controls and the requirement to maintain reliability under rapid technological change.

Agentic-commerce use cases include authenticated procurement of mission components, controlled maintenance and cross-organizational logistics. Spherity could add supplier identity, product and component credentials, bounded service mandates, PQC-resilient data-sharing corridors and lifecycle receipts. Its roadmap could further support mission-data entitlements, run-time trust checks, dynamic revocation and signed incident evidence across multi-organization deployments.

7.9 Sereact: Physical AI for flexible robotic handling

Stuttgart-based Sereact develops AI that lets robots perceive and manipulate unfamiliar objects without extensive task-specific programming. In July 2026 it announced a \$116 million Series B, more than 200 live systems and over one billion production picks. Its Cortex platform aims to make Physical AI reusable across warehouse and production environments.

The defensible advantage is production data and deployment experience in robotic manipulation. The catalyst is demand for flexible automation in environments with changing products and labour constraints. The key risk is robust generalization: a system that performs well in one warehouse must remain safe and reliable across edge cases, hardware types and operating conditions.

In agentic commerce, a handling system can trigger replenishment, verify an item or coordinate a service. Spherity could provide robot identity, trusted product passports, component evidence, service-provider credentials, data provenance and authorization for external actions. Future runtime integrity checks, safety-state attestations and dynamic revocation could add continuous assurance without replacing the robot's certified safety functions.

7.10 Spherity: the cross-layer trust fabric

Spherity develops digital identity and compliance infrastructure for organizations, products and regulated data exchange. Founded in 2017 and headquartered in Dortmund, the company reports a team of 30 people across eight remote locations. Its current product foundations include CARO for US pharmaceutical supply-chain credentialing, VERA for Digital Product Passports and Digital Battery Passports, and EIDA for organizational identity and wallet-based data sharing.

Spherity's importance does not rest on being the largest company in the list. It rests on structural position. The other nine companies make agents more capable, processes more intelligent and machines more autonomous. Spherity helps make actions portable across organizational boundaries by connecting who is acting, what they may do and which evidence supports the decision.

The five-layer stack makes that position concrete. At Layer 1, Spherity's verifiable-credential and crypto-agility foundations can connect to roadmap integrations with secure elements, HSMS, hardware attestation and PQC migration. At Layer 2, it connects legal-entity, representative and agent identity with European Business Wallets, mandates and signed intent declarations. At Layer 3, it links DPP, DBP and future AI System Passport evidence through provenance and lineage. Layers 4 and 5 define the strategic roadmap: machine, component and software identity; BOM and SBOM graphs; configuration and safety-state attestations; runtime integrity and policy checks; dynamic authorization; monitoring; revocation; and incident evidence. These roadmap functions would surround and interoperate with certified safety and operational systems rather than replace them.

The market pull is unusually broad. The EU Ecodesign for Sustainable Products Regulation requires product-passport information to be based on open standards and be interoperable, machine-readable, structured, searchable and transferable without vendor lock-in. The EU Battery Regulation requires passports from 18 February 2027 for specified categories including electric-vehicle and large industrial batteries. The European Commission's proposed European Business Wallet framework aims to give businesses a secure, interoperable identity and data-sharing instrument. The AI Act, Cyber Resilience Act and EU PQC roadmap add further pressure for governed AI, secure products and crypto-agility.

This supports recognized indicators of a high-quality technology opportunity without relying on a fundraising narrative: large cross-sector markets; regulatory and technological tailwinds; specialized identity and compliance expertise; reusable software; partner-led distribution; recurring enterprise workflows; international expansion; and the possibility of becoming embedded in processes that are costly to replace. Standards-based deployment can also create network effects: each additional credential issuer, verifier, supplier, product or agent makes the trust network more useful.

The risks must be stated just as clearly. Regulatory markets can develop more slowly than expected. Standards can fragment. Large enterprise platforms may bundle parts of the trust layer. Adoption requires ecosystem coordination, not only good software. Spherity must therefore demonstrate repeatable deployments, reduce integration effort and convert regulatory deadlines into high-value operational and commercial use cases.

Its international evidence is tangible but should not be overstated. Spherity operates from Germany and Europe, supports US pharmaceutical compliance through CARO and a New York presence, signed a 2025 memorandum with South Korea’s K4Security, and publishes China-facing Digital Product Passport material. A specific Japanese office or subsidiary was not verified for this article; Japan should be described as a priority market only when supported by additional evidence.

8 From regulatory compliance to transactive intelligence: Spherity’s strategy

Spherity’s strategy begins with regulated, near-term problems and expands toward trusted autonomy.

First, strengthen the cryptographic and hardware-rooted trust foundation. Verifiable credentials and crypto-agility are current architectural foundations. PQC migration, secure-element and HSM integration, and hardware attestation extend that foundation with semiconductor, cloud and security partners. The objective is portable trust that can survive infrastructure changes and long asset lifecycles.

Second, deploy European Business Wallets, verifiable semantic trust chains and Digital Product Passport applications. Business wallets establish legal-entity identity, representation and reusable company credentials; product and battery passports provide structured lifecycle evidence for products and assets. Together, they address immediate compliance and onboarding needs while creating the identity and evidence base for agents.

Third, converge organizational, agent and product identity. A trustworthy product claim needs an accountable issuer. A company credential becomes more useful when connected to the representatives, agents, products, facilities and machines for which the organization is responsible. This convergence lets a system verify both *what the component is, who stands behind the evidence* and *which agent may act on it*.

Fourth, make evidence AI-readable and verifiable. Industrial agents need structured claims, issuer status, provenance, timestamps, revocation, policy and links between evidence—not only PDFs and webpages. DPPs, DBPs and future AI System Passports can feed evidence graphs that show which credentials, BOM or SBOM relationships, configurations and transformations support a conclusion.

Fifth, give AI agents bounded and verifiable authority. An agent may represent a purchasing department, machine owner, energy asset or public authority. Its signed mandate and intent declaration should specify scope, duration, counterparty, value or operational limit,

product category, jurisdiction and escalation rules. European Business Wallets can connect the agent to the legal entity and representative responsible for it.

Sixth, extend trust into Physical AI and runtime assurance. The roadmap adds machine, component and software identity; BOM and SBOM graphs; configuration and safety-state attestations; and runtime integrity checks. Spherity would verify trust conditions and preserve evidence, while certified controllers and safety systems retain responsibility for physical execution.

Seventh, build continuous mission assurance. At the highest layer, the control and data planes support runtime trust and safety checks, dynamic authorization, safe-state policies, monitoring, revocation and incident evidence. Automated responses can include allow, deny, restrict, escalate, require multi-party approval or request a safe-state transition through an approved operational system. This is a strategic roadmap, not a claim that Spherity currently replaces functional-safety or mission-control platforms.

Eighth, enable transactive digital twins and agentic Physical AI. A digital twin becomes transactive when it can use trusted lifecycle and runtime evidence, act through an authorized agent and enter a governed exchange. A machine then becomes more than a data source: it can request service, buy an approved replacement, sell capacity or prove compliance while keeping humans and organizations accountable.

This progression explains why compliance infrastructure can become commercial infrastructure. The same credentials used to satisfy a regulator can shorten supplier onboarding. The same product passport used for market access can help an agent select compatible parts. The same organization wallet used to prove representation can authorize an agent to transact. Compliance is the beachhead; lower-friction, higher-trust commerce is the expanding value pool.

9 Seven examples of trusted agentic commerce

1. **Factory maintenance.** A factory agent detects a failing bearing, checks compatible replacements in Digital Product Passports, verifies the supplier's business credential and places an order within an approved budget. The receipt records evidence, mandate, product and supplier status.
2. **Robot service.** A robot requests maintenance from a credentialed technician. The service provider proves authorization; completed work and installed parts are added to trusted lifecycle evidence.
3. **Battery flexibility.** A battery agent proves asset identity, ownership, capacity and state of health before offering flexibility to an energy market. The owner's mandate limits price, cycle depth and trading period.
4. **Sustainable procurement.** A procurement agent evaluates origin, carbon data, repairability, certifications and supplier eligibility. It escalates conflicting evidence and contracts only when policy conditions are met.
5. **Cross-border logistics.** A logistics agent verifies carrier credentials, insurance, shipment conditions and customs documents before booking. Status and revocation are checked at execution, not only at onboarding.
6. **Consumer purchasing.** A consumer agent compares durability, warranty, authenticity and repair data from verified passports rather than relying only on marketplace descriptions.
7. **Public procurement.** A government agent verifies supplier eligibility, licenses, sanctions and product compliance before initiating a transaction, while keeping a complete evidence trail

for review.

10 Why collaboration with Spherity creates mutual value

The collaboration thesis is specific: Spherity adds the organizational authority, product evidence and durable audit layer that lets partners extend automation beyond their own boundaries.

- **Infineon + Spherity:** connect secure elements, HSMs and chip-rooted identifiers to hardware attestation and organizational, component and agent credentials; build PQC-ready chains from silicon to transaction.
- **Schaeffler + Spherity:** create verified actuator and bearing passports, BOM relationships and configuration attestations; authorize robot maintenance and replacement orders; preserve runtime and service evidence.
- **Phoenix Contact + Spherity:** authenticate controllers, configurations, software updates and agent identities; evaluate runtime trust conditions and request approved restrictions or safe-state transitions without bypassing certified control.
- **Celonis or SAP + Spherity:** add portable organizational authority and external evidence provenance to enterprise workflows and process intelligence.
- **n8n or Parloa + Spherity:** verify agent identity and delegated authority before a workflow or conversational agent commits to a transaction; issue an action receipt afterward.
- **NEURA or Sereact + Spherity:** combine robot identity, component and software evidence, technician credentials, configuration and safety-state attestations, runtime integrity checks and dynamic revocation.
- **1KOMMA5° + Spherity:** link asset passports, installer credentials, grid permissions and trading mandates for distributed energy.
- **Helsing or Quantum Systems + Spherity:** establish PQC-resilient supplier, mission and data-sharing corridors with auditable entitlements, runtime trust checks, dynamic authorization and signed incident evidence.

Across B2B, sensitive B2C, B2B2C, B2G, G2G and machine-to-machine processes, the commercial benefits are similar: faster onboarding, lower verification and compliance cost, reduced fraud, safer automation, access to regulated markets, less transaction friction and new service revenues. A passport can support predictive maintenance; a wallet can support reusable supplier onboarding; an action receipt can support insurance, dispute resolution and outcome-based contracting.

11 Global digital data-sharing and commerce corridors

Agentic industry is inherently cross-border. A German machine can include Asian components, operate in a US-regulated supply chain and be serviced by a European provider. Trust therefore cannot stop at a national identifier or one platform account.

An EU–US or EU–APAC corridor needs mutual recognition of organizational credentials, product and asset evidence, importer and manufacturer identity, agent mandates and transaction receipts. Regulatory differences will remain, but common technical patterns can let a verifier understand who issued a claim, which jurisdictional rules apply, whether the credential is current and what evidence must be retained.

Spherity’s CARO experience in the US pharmaceutical supply chain provides a practical base: regulated trading partners must establish organizational legitimacy and authorized status. Its South Korean K4Security partnership creates a route for European Digital Product Passport readiness in Korea and the wider APAC market. China-facing materials address one of the world’s most consequential manufacturing ecosystems. These are useful starting points for reciprocal digital corridors; they are not by themselves proof that legal recognition is solved.

PQC matters because industrial assets, product passports, pharmaceutical records and infrastructure agreements can remain sensitive or legally relevant for many years. NIST finalized its first three PQC standards in 2024, while the European Commission published a coordinated implementation roadmap in 2025. The practical strategy is crypto-agility: inventory dependencies, separate trust semantics from algorithms and enable controlled migration without invalidating the business meaning of identity, authority or evidence.

12 An aggressive roadmap for German industrial competitiveness

Germany should treat trusted agentic cyber-physical systems as shared industrial infrastructure, comparable to a standards-based logistics or payment network. The public role is not to select one platform winner. It is to define interoperable rules, fund production corridors, use procurement and measure outcomes.

12.1 2026–2027: establish the trust foundation

Germany should publish a national reference architecture for legal-entity, representative, machine and agent identity; align European Business Wallets with Digital Product Passports and Digital Battery Passports; define the control and data planes; inventory PQC dependencies; establish profiles for secure-element, HSM and hardware-attestation integration; and launch lighthouse corridors in manufacturing, energy, health supply chains and critical infrastructure.

KPIs: 10,000 credentialed organizations; one million product or asset credentials; 100 production agent credentials; onboarding time below one day for a previously verified SME; at least five production-grade DPP–wallet corridors; PQC inventories covering 80% of participating critical systems.

12.2 2027–2028: move from pilots into production

Regulators and industry should issue machine-readable permits and company or professional credentials; define agent-authorization, signed-intent and action-receipt profiles; establish an interoperable AI System Passport concept; integrate with Manufacturing-X, Catena-X and European data spaces; and provide packaged onboarding for SMEs. Sandboxes should test procurement, logistics and energy transactions with real economic value under capped mandates and continuous credential-status checks.

KPIs: 100,000 credentialed organizations; ten million products or assets; 10,000 agents; at least 25 production corridors; 50% reduction in repeated onboarding checks; 25% SME participation; measurable unauthorized-action rate below 0.1% with mandatory incident reporting.

12.3 2028–2029: scale agentic industrial markets

The focus should shift to machine-to-machine transactions, transactive twins, shared conformance evidence and cybersecure, functionally safe agent certification. Germany should enable machine, component and software identity; verifiable BOM and SBOM graphs; configuration and safety-state attestations; runtime integrity and policy checks; dynamic authorization; and monitored autonomous procurement, service and energy-market actions. Trust services should interoperate with certified controllers rather than duplicate their physical safety functions.

KPIs: €10 billion in annual agent-mediated transaction value under verifiable mandates; 100 million credentialed products or assets; 100,000 authorized agents and machines; 30% productivity improvement in targeted verification workflows; 50% reduction in compliance handling time; 50% of corridor cryptography migrated or demonstrably ready for approved PQC profiles.

12.4 2029–2031: export the European trust stack

Germany and the EU should establish reciprocal US and APAC trust corridors, recognize organizational, agent, product and operational-assurance credentials across defined schemes and publish exportable reference architectures for industrial agentic commerce and continuous mission assurance. Procurement and trade policy should reward interoperable, auditable and crypto-agile implementations with tested revocation, incident evidence and recovery orchestration.

KPIs: at least six live international corridors; one million participating organizations; one billion credentialed products, assets or components; €100 billion in annual agent-mediated transaction value; onboarding measured in minutes for recognized credentials; 90% PQC readiness for long-lived critical trust services; material year-on-year reduction in fraud and unauthorized actions.

These targets are intentionally aggressive. The point is to measure industrial adoption and trust outcomes—not the number of workshops, strategies or isolated pilots.

13 German Technology Outlook 2030

13.1 Scenario 1: fragmentation

Germany continues to produce excellent companies, but identity, data, product and agent systems remain incompatible. Corporates run pilots inside their own platforms; SMEs repeatedly complete the same onboarding; machines can recommend actions but cross-company transactions require manual intervention. Value accrues slowly because every connection is bespoke.

13.2 Scenario 2: platform dependence

German industry adopts capable foreign AI and commerce platforms without retaining control over the trust, data and transaction layer. Deployment may be fast, but strategic dependency grows. Industrial data, agent governance and market access become tied to a small number of external ecosystems. Germany remains an equipment supplier while others control the software-defined transaction layer.

13.3 Scenario 3: coordinated industrial stack

Germany combines sovereign compute, enterprise software, industrial data, Physical AI and European trust architecture through open standards. Companies compete at their own layer and interoperate where shared trust produces more value. Agents can discover services, validate counterparties and product evidence, act within bounded mandates and produce receipts that regulators, insurers and business partners can understand.

The resulting economy supports autonomous but accountable factories, trusted procurement agents, machine-to-machine services, secure energy markets, transactive digital twins, cross-border compliance automation and PQC-resilient trade corridors. It also gives German ventures a larger home market and corporate anchors a faster route to new services.

The coordinated scenario does not advocate a closed national technology stack. It proposes an open European strategy that connects German strengths with complementary capabilities across foundation models, sovereign cloud, industrial connectivity, secure edge computing, virtual twins, automation, autonomous mobility and mission systems. Europe's competitive advantage will not come from one national champion, but from making these capabilities globally interoperable and demonstrably cybersecure, functionally safe, compliant and accountable in the physical economy.

Germany already has the components. The next task is institutional and commercial: connect category-creating ventures, industrial anchors, public procurement, standards bodies and research institutions around real production corridors. In that system, Spherity's most credible ambition is not to displace the other companies. It is to become the trust fabric that helps them transact with more partners, in more regulated markets, with less friction and clearer accountability.

14 Frequently asked questions

This section provides concise answers to the strategic, operational and technology questions raised by the analysis.

14.1 Strategic positioning

14.1.1 How should Germany and Europe position themselves geopolitically, and compete in the global AI race?

Germany and Europe should compete where they possess structural advantages: advanced industry, engineering excellence, regulated markets, functional safety, cybersecurity and trusted cross-border commerce. The strategic objective should not be to replicate every global foundation model or create a closed European technology island. It should be to establish the world's most interoperable and trustworthy environment for deploying AI agents, machines and autonomous systems in the physical economy. Open standards, coordinated procurement and internationally recognized trust corridors can connect European infrastructure with leading models and markets in the US and APAC while preserving European control over sensitive data, operational rules and industrial value creation.

14.1.2 What is a model-agnostic Digital Sovereignty Stack?

A model-agnostic Digital Sovereignty Stack allows organizations to select, combine or replace European, international, open-source and specialized AI models without rebuilding the surrounding identity, data, governance, security and operational infrastructure. It connects compute and connectivity, AI models and agent orchestration, industrial data and digital twins, verifiable credentials, European Business Wallets, product and AI-system evidence, authorization and runtime assurance through open standards. Digital sovereignty therefore does not mean technological isolation. It means retaining control, portability and strategic choice while preserving trusted identities, evidence provenance, policy enforcement, auditability and operational continuity when models, cloud providers or technology partners change.

14.1.3 Why can trust infrastructure become Europe’s and Germany’s distinctive advantage?

As AI models become more capable and interchangeable, the critical challenge moves from generating an answer to authorizing and governing actions across organizational and physical boundaries. Europe’s and Germany’s potential advantage is a cross-company trust infrastructure connecting legal-entity, representative, agent, machine and component identities with verifiable mandates, intent declarations, product and AI-system evidence, hardware and software attestations, crypto-agility and post-quantum security. Combined with runtime policy checks, dynamic authorization, monitoring, revocation and incident evidence, this infrastructure enables secure, compliant, risk-controlled and accountable AI deployment. It complements—but does not replace—cybersecurity platforms, governance controls, certified functional-safety systems or human responsibility.

14.1.4 Which German companies are shaping Industrial AI, Physical AI and trusted agentic systems?

The selected German Top 10—1KOMMA5°, Black Forest Labs, Celonis, Helsing, n8n, NEURA Robotics, Parloa, Quantum Systems, Sereact and Spherity—represent complementary capabilities spanning energy platforms, AI models, process intelligence, autonomous systems, orchestration, robotics, conversational AI and trust infrastructure. Industrial anchors including Bosch, Deutsche Telekom, Festo, Infineon, Phoenix Contact, SAP, Schaeffler, Schwarz Digits, Siemens and TRUMPF provide production environments, industrial technology, infrastructure and global distribution. The strategic opportunity lies less in ranking these companies against one another than in combining their capabilities into an interoperable German and European technology stack.

14.1.5 Which non-German European companies best complement the agentic cyber-physical technology stack?

A strategically balanced non-German European Top 10—selected for its fit with the five-layer stack rather than size alone—is: ABB for robotics and Physical AI; Dassault Systèmes for virtual twins and Industrial AI; Leonardo for cybersecure mission systems; Mistral AI for foundation models; Nokia for industrial connectivity and edge infrastructure; NXP Semiconductors for secure edge computing; OVHcloud for sovereign cloud; Schneider Electric for energy technology and open automation; STMicroelectronics for embedded AI and hardware-rooted security; and Volvo Group for connected and autonomous commercial mobility. Together, they complement the German stack across compute, models, data, automation, mobility and mission

systems. Spherity can add interoperable identity, evidence, mandates, runtime authorization and accountable cross-company transactions.

14.2 Deployment into sensitive and critical operations

14.2.1 What must be demonstrated before Industrial AI or Physical AI is deployed in sensitive or critical operations?

Before an AI system is connected to production assets, operational decisions or physical processes, its operators should demonstrate who developed, supplied, configured and authorized it; which data, software, components and evidence support its decisions; who remains accountable; and within which operating boundaries it may act. Deployment readiness should combine cybersecurity assessment, functional-safety analysis, regulatory compliance, operational-risk limits, configuration and safety-state attestations, human-oversight arrangements and tested fallback responses. These requirements are particularly important in regulated industries, critical infrastructure, defence and other environments where an incorrect, manipulated or unauthorized action could cause physical, economic or societal harm.

14.2.2 Why must runtime assurance and control automation be designed before deployment?

Pre-deployment testing cannot anticipate every change in operating conditions, system configuration, threat exposure, credential status, model behaviour or evidence quality. Critical deployments therefore require runtime assurance that continuously checks agent identity and authority, software and configuration integrity, evidence freshness, cybersecurity conditions, compliance rules, risk thresholds and the current safety state. Pre-approved control automation should be able to authorize, restrict or block an action; require human escalation; revoke compromised credentials; initiate an approved safe-state response; and preserve action and incident evidence. These controls must be designed, governed and tested before deployment, while certified safety controllers and accountable human authorities retain responsibility for physical intervention.

14.2.3 How can trusted AI infrastructure support defence and civil–military supply chains?

Defence, dual-use and civil–military supply chains require organizations to verify suppliers, personnel, agents, machines, components, software and operational evidence across security domains and jurisdictions without unnecessarily disclosing sensitive information. Verifiable credentials, selective disclosure, hardware-rooted identity, BOM and SBOM provenance, product and AI-system evidence, bounded agent mandates, post-quantum-secure trust chains, continuous status checks and revocation can strengthen supply-chain assurance and make security, eligibility, export-control and compliance decisions more machine-verifiable. This infrastructure can enable faster and more secure collaboration across defence and civilian ecosystems, but it does not replace classified networks, security accreditation, certified safety systems, command authority or national security controls.

14.3 Agentic commerce and the physical economy

14.3.1 What is agentic commerce?

Agentic commerce is commerce in which AI agents discover, evaluate, negotiate, order, coordinate and document transactions on behalf of people, organizations, machines or public authorities. Unlike conventional automation, an agent may select counterparties, interpret evidence and initiate actions within a delegated mandate. Payment is only one part of the process. Trusted agentic commerce also requires verifiable identity, authority, product and compliance evidence, policy enforcement and an accountable record of what the agent decided, which evidence it used and who remains responsible for the outcome.

14.3.2 What is industrial agentic commerce?

Industrial agentic commerce applies AI agents to procurement, maintenance, logistics, energy, capacity, spare parts and regulated services. Transactions are frequently machine-to-machine or B2B and can affect production assets or physical operations. An industrial agent must therefore respect technical compatibility, safety rules, approved suppliers, budgets, jurisdictional requirements and operating constraints. It should be able to verify the supplier, product, mandate and relevant evidence before acting and produce a machine-verifiable receipt documenting the decision, authorization, transaction and resulting operational state.

14.3.3 How is agentic commerce different from e-commerce?

E-commerce digitizes a transaction interface primarily for a human buyer who remains responsible for evaluating the offer and approving checkout. Agentic commerce delegates parts of discovery, evaluation, negotiation and execution to software agents that may operate continuously and across multiple platforms. This requires more than a user account, API key and payment method. The agent needs a verifiable principal, a bounded mandate, trustworthy product and counterparty evidence, machine-enforceable policies and a durable record of its actions. The principal difference is therefore not automated payment but delegated, evidence-based and accountable decision-making.

14.3.4 What infrastructure do AI agents need before they can transact?

AI agents need a control plane and a data plane. The control plane establishes buyer, supplier, representative and agent identity; mandates, intent declarations and authorization; runtime policy decisions; revocation; and action receipts. The data plane supplies trusted product, software, asset and compliance evidence with verifiable provenance and status. Physical and mission-critical systems additionally require machine and component identity, configuration and safety-state attestations, runtime integrity checks, monitoring and integration with approved control responses. Long-lived transactions and evidence chains also require hardware-rooted trust, crypto-agility, post-quantum migration planning and records for audit and dispute resolution.

14.4 Trust infrastructure building blocks

14.4.1 Why do AI agents require verifiable organizational identity?

A counterparty must know which legal entity stands behind an AI agent, who is authorized to represent that entity and whether the agent may bind it for the proposed action. A model name,

API key or platform account can identify a technical service but does not prove legal representation or delegated authority. Verifiable organizational, representative and agent credentials can connect the agent to an accountable principal, define the scope and duration of its mandate and allow counterparties to check status or revocation. This makes organizational identity a foundation for enforceable contracts, regulated transactions and cross-company accountability.

14.4.2 How can European Business Wallets support agentic commerce?

European Business Wallets are intended to provide companies with secure, interoperable ways to identify themselves and exchange authenticated business information. Connected to representative and agent mandates, a wallet could make organizational credentials, licences, authorizations and other verified attributes reusable across transactions. An agent could then demonstrate which company it represents and which actions it may perform without repeating the complete onboarding process for every counterparty. This could reduce verification costs and support cross-border agentic commerce, although the European Business Wallet initiative remained a legislative proposal at the research cut-off.

14.4.3 How do Digital Product Passports support AI-agent decisions?

Digital Product Passports can provide structured information about a product’s identity, composition, origin, lifecycle, repairability and compliance status. AI agents can use this evidence to compare products, determine technical compatibility, confirm regulatory eligibility and select approved suppliers. A passport alone, however, does not make every claim trustworthy. The agent must also verify who issued the evidence, whether it is current, whether relevant credentials remain valid and how the information was derived. Digital Product Passports become most valuable for agentic decisions when combined with verifiable provenance, status checks and accountable organizational identity.

14.4.4 Why is post-quantum cryptography relevant to trusted AI and agentic commerce?

Industrial identities, product records, contractual evidence and infrastructure agreements may need to remain trustworthy for many years. Future quantum computers could weaken widely used public-key algorithms, creating risks for long-lived data and trust relationships. Post-quantum cryptography provides replacement algorithms, while crypto-agility allows organizations to change algorithms and keys without redesigning every identity, compliance or transaction process. A practical strategy begins with identifying cryptographic dependencies, separating trust semantics from individual algorithms and planning controlled migration across hardware, credentials, wallets, products and cross-border data-sharing corridors.

14.5 Spherity’s role

14.5.1 What role can Spherity play in trusted AI and agentic commerce?

Spherity can provide complementary trust and compliance infrastructure across the stack. Current foundations include verifiable credentials, legal-entity and agent identity, European Business Wallets, product and AI-system evidence, provenance, mandates, authorization, action receipts and crypto-agility. The roadmap extends toward hardware attestation, machine and

software identity, configuration and safety-state attestations, runtime integrity, dynamic authorization, monitoring, revocation and incident evidence. These capabilities support AI models, cloud platforms, robots, enterprise applications and certified controllers without replacing them.

14.5.2 Why Spherity?

Most technology providers optimize within their own platform boundary. Spherity’s strategic value is to provide a model- and platform-agnostic trust layer across those boundaries, so partners can verify counterparties, reuse evidence, enforce delegated authority and preserve accountability without surrendering control of their core systems. This adjacent position can reduce onboarding and compliance friction, support access to regulated markets and enable governed automation across company and national borders. Spherity complements—not replaces—cybersecurity platforms, certified functional-safety controls, mission systems or human responsibility.

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Fact-check caveats: The Cohere–Aleph Alpha transaction is described as planned, not closed. The European Business Wallet is described as a Commission proposal, not adopted law. Company forecasts, such as NEURA’s robot targets, 1KOMMA5°’s controlled-capacity target and roadmap KPIs, are labeled as company targets or forward-looking proposals. CARIAD is not selected; it remains active after restructuring, while the Bosch–CARIAD Automated Driving Alliance has concluded.

16 Top 10 company profile table (in alphabetical order)

Investor-readable comparison of strategic opportunity, scalability, execution risk, agentic-commerce relevance and the complementary Spherity fit.

Company	Technology category	Defensible advantage	Principal catalyst	Scalability and capital profile	Principal risk	Agentic-commerce relevance	Potential Spherity collaboration
1KOMMA5° / Heartbeat AI	Distributed-energy orchestration	Installed asset base, customer access and market integration	Electrification, volatility and flexibility markets	Platform/network effects with hardware-heavy execution	Regulation, hardware heterogeneity and operational responsibility	Asset agents trade energy and flexibility	Asset passports, installer credentials, grid permissions and mandates
Black Forest Labs	Visual foundation models	Frontier visual-model expertise and developer adoption	Multimodal enterprise and creative demand	High software leverage but compute-intensive R&D	Model competition, copyright and provenance	Visual inspection and content agents supporting transactions	Model identity, AI System Passport evidence, provenance and usage-policy receipts
Celonis	Process intelligence	Cross-system process context and enterprise adoption	Agents moving from chat to execution	High recurring software potential; integration-heavy enterprise sales	Platform competition and difficult value realization	Detect and orchestrate cross-company procurement actions	External credentials, product evidence, mandate checks and receipts
Helsing	Defence AI and mission autonomy	Operational software, domain integration and European ownership	European defence modernization	Software leverage with capital-intensive systems integration	Qualification cycles, geopolitics, deployment assurance	Secure supplier, maintenance and mission-support transactions	PQC-ready supplier identity, mission-data entitlements, runtime trust checks and incident evidence
n8n	Workflow and agent orchestration	Flexible, developer-friendly integrations and controlled deployment	Enterprise agent adoption	High software scalability and ecosystem distribution	Crowded automation market; governance burden	Orchestrates discovery, verification, policy and execution	Verify agent mandates, intent declarations, counterparties and evidence before tools run
NEURA Robotics	Cognitive and humanoid robotics	Full-stack robot, sensor, control and ecosystem integration	Labour scarcity and flexible automation	Large market with hardware and production capital needs	Reliability, safety, cost and manufacturing scale	Robots procure service or parts within bounded mandates	Robot/software identity, configuration and safety-state attestations, runtime integrity and receipts
Parloa	Enterprise conversational AI	Voice and enterprise customer-service specialization	Contact-centre shift to agentic operations	Scalable enterprise software with model and integration costs	Model commoditization and erroneous commitments	Orders, service changes, refunds and negotiated outcomes	Representative and agent identity, intent declarations, limits and action receipts
Quantum Systems	Autonomous systems	Integrated platforms, sensing and autonomy software	Defence and dual-use autonomous-system demand	Scalable software plus manufacturing requirements	Production, procurement concentration and export controls	Authenticated parts, maintenance and logistics	Supplier/product credentials, mission entitlements, runtime trust checks and PQC corridors
Sereact	Physical AI for robotic handling	Production data and generalizable manipulation deployments	Flexible warehouse and factory automation	Repeatable software plus hardware integration	Edge-case reliability and multi-environment generalization	Replenishment, item verification and autonomous service triggers	Robot identity, BOM evidence, safety-state attestations, runtime integrity and dynamic revocation
Spherity	Cross-company trust and continuous-assurance infrastructure	Convergence of legal-entity and agent authority, trusted evidence and runtime assurance	EBWs, DPP/DBP, AI governance, CRA, cyber-physical autonomy and PQC migration	Capital-efficient software; partner distribution; cross-sector reuse	Standards fragmentation, ecosystem coordination, certification boundaries and platform bundling	Provides identity, authority, evidence, runtime policy inputs and audit beneath transactions	Native cross-layer role: current foundations plus hardware, Physical AI and mission-assurance roadmap without replacing certified control